A reading of known relations among the constants: the Standard-Model strong-to-weak ratio at the Planck scale, 1.0768, as a ratio of units, and Koide's lepton relation as a balance between the same and the different

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Abstract

Run to the Planck mass with the Standard Model at two loops, and with no new particles in between, the measured gauge couplings (PDG 2025, CODATA 2022) give $1/\alpha_3 = 52.98 \pm 0.07$ and $1/\alpha_2 = 49.200 \pm 0.008$ at M_P. Their ratio is $R = 1.0768 \pm 0.0014$ (1.0766 with the formulas of Buttazzo et al. 2013): a Standard-Model number, not a new result. This record proposes one reading of known facts. Inverse couplings are read as counts of readings per elementary cell; the inverse weak coupling as the count in the unit of time, the inverse strong coupling as the count in the unit of space, with counts inversely proportional to the unit. R is then the original ratio between the two units, taken as an initial datum.

How strong are the forces of nature? A ratio between two forces read as a ratio between units of space and of time, and the masses of particles as a balance between the same and the different

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In short: what you find in this text

In nature there are four forces. Three act between particles: the strong force, the weak force and the electromagnetic force. The fourth is gravity. Each force has a number that says how intense it is. This number is called the **coupling**. It is written with the Greek letter α (alpha). A large coupling means an intense force. Physicists often use its inverse, 1/α. A large 1/α means a weak force.

The couplings are not fixed. They change with the energy at which the particles are observed. One says that the couplings *run*. For the electromagnetic force you can imagine it like this: a charge is surrounded by a kind of cloud that partly hides it. The closer you get, that is, the higher the energy, the less cloud remains in between, and the force appears more intense. For the strong force the opposite happens: up close it appears weaker. It is not an illusion: the coupling really changes with energy.

The **Standard Model** is the theory that describes the particles and the three forces between them. With this theory one can calculate how the couplings run. The calculation is done in degrees of precision, called **loops**. One loop is a first calculation. Two loops are a more precise calculation.

We carry the couplings up to the **Planck mass**. It is an enormous energy, the one where gravity becomes as strong as the

other forces. We use the Standard Model at two loops. We assume that there are no new particles in between. We start from the measured couplings (PDG 2025, CODATA 2022). The PDG and CODATA are the official collections of measured values in physics. The result is:

$1/\alpha_3(M_P) = 52.98 \pm 0.07$, $1/\alpha_2(M_P) = 49.200 \pm 0.008$, $\mathbf{R = 1/\alpha_3 \div 1/\alpha_2 = 1.0768 \pm 0.0014.}$

Here α_3 is the strong coupling and α_2 the weak one. M_P is the Planck mass. R is the ratio between the two inverses. The ± sign gives the uncertainty: how much the number can be off.

This is a Standard-Model number. It is not a new result. Buttazzo and colleagues (2013) give 1.0766 with the same inputs. Everything up to here is standard physics. The structural facts used in this record are standard physics too. There are four:

- the electromagnetic coupling is a mixture of the weak coupling and the hypercharge coupling (hypercharge is another charge of the particles);
- the running depends on the number of *colours* and on the particles lighter than the energy scale;
- charges weigh with their square;
- there is a link between the strong coupling at the Planck scale and the gravitational weakness of the proton (Wilczek).

The record proposes a **reading** of these known facts. A reading is a way of interpreting them. The inverses of the couplings are read as **counts of readings per elementary cell**. The cell is the smallest piece. A reading is an elementary event that the cell registers. The inverse of the weak coupling is the count in the **unit of time**. The inverse of the strong coupling is the count in the **unit of space**. The counts are inversely proportional to the unit: small unit, large count. Then R is the original ratio between the two units. It is taken as an initial datum.

The reading is put to the test with gravity, in three forms registered before the calculation. First form and second form: R as a real physical difference between space and time. Then R would give a post-Newtonian parameter $\gamma = 2R/(1+R)$ or $2/(1+R)$. The parameter γ says how much a mass curves space. How much gravity bends and slows down light depends on γ: the Cassini probe measured it with the delay of its radio signals. These values lie about 1600 times the error of the Cassini measurement away. If the working uncertainty of R is included, they lie about 59 times the combined error away. These two readings fall. Third form: R as a pure ratio of units. This gives $\gamma = \beta = 1$: it is compatible with the Cassini measurement. With R and the measured weak coupling, the strong coupling at the Planck scale follows. And with the known running so does $\alpha_s(M_Z)$, the strong coupling at the energy of the Z particle.

Here **no constant is derived**. Before the calculation twenty-eight mechanisms were registered. Each was a way of obtaining the strong coupling from the model's counts. None reproduces $\alpha_s(M_Z)$ within two standard deviations. By pure chance 0.22 hits would be expected.

There is a second part, **Part B**. It reads the lepton masses in the same spirit. The leptons are a group of particles: the electron, the muon, the tau and the three neutrinos. Three particles with the same charge form a *family*: the electron, the muon and the tau are one family, the three neutrinos another. The three particles of a family are called *generations*: they are similar, but they have different masses. The mass term of a family has **amplitudes**. These amplitudes split into two parts. One part is shared by the three generations: **the same**. The other part distinguishes them: **the different**. The proposed reading is that the two parts weigh the same.

This is exactly **Koide's relation**. The charged-lepton pole masses satisfy it within their measurement uncertainty. The pole mass is the mass of the free particle, as if one could weigh it alone. The uncertainty is about eight parts in a million. The central value lies three parts in a million from 2/3.

Applied to neutrinos, the balance has one solution for each ordering of the masses. The neutrino masses can be ordered in two ways: normal or inverted. There is a cosmological bound: in the ΛCDM model it is 0.077 eV (Planck + ACT + DESI DR2, PDG 2025). With this bound only the solution with normal ordering remains. The sum of the neutrino masses is 0.0593 ± 0.0003 eV. In another model, w_0w_aCDM, the bound is 0.186 eV, and both solutions remain.

The same relation had already been applied to neutrinos by Brannen (2006). With the data of the time he found a sum of 0.0600 eV. The reading adds a **proposed reason**, an interpretation. It does not add a new number.

For the quarks things go like this. Three rules were registered on how the *colour* of the quarks shifts the balance. None of the three holds. Then the quarks were read *in the mirror*, that is, with the inverse masses. In this way the down-type quarks satisfy the balance, within their uncertainties. This relation was already known in the literature. The up-type quarks instead do not satisfy it.

How to read this record

The order is this. First the measured data. Then a direction, declared as a hypothesis. Then where the numbers meet. Only at the end the reading proposed by the model. For each group of constants that seem to meet, two questions are asked:

1. do these constants meet today, without the hypothesis?
2. are there different hypotheses that make them meet?

Part A (sections 1–8) concerns the couplings of the forces. Sections 1–4 are standard physics. Section 5 is the proposed reading. Sections 6–8 say what is known, what was tried and what is not claimed. **Part B** (sections B and B.2) concerns the masses of the leptons and of the quarks, in the same order.

Every computed number comes from the code of the record. The measured data, the assigned estimates and the values taken from the literature are marked as such. An example of an assigned estimate is the 0.02 of section 2.

1. The data

All values come from CODATA 2022 and from the Review of Particle Physics of the PDG. In the table the numbers in brackets are the uncertainty on the last digits. For example 137.035999177(21) means that the last two digits can vary by about 21. *MS-bar* is a convention for defining quantities. It is like deciding to always measure a person's height without shoes.

quantity	value	source
$1/\alpha(0)$	137.035999177(21)	CODATA 2022
$1/\alpha^{(5)}(M_Z)$, MS-bar	127.930 ± 0.008	PDG electroweak review
$\sin^2\theta_W(M_Z)$, MS-bar	0.23122 ± 0.00006	PDG electroweak review
$\alpha_s(M_Z)$	0.1180 ± 0.0009	PDG QCD review, world average
M_Z, M_W, M_H	91.1880(20), 80.3692(133), 125.20(11) GeV	PDG summary tables
m_t (direct), m_t(m_t) MS-bar	172.56 ± 0.31, 162.5 (+2.1 −1.5) GeV	PDG summary tables
m_e, m_p	0.51099895069(16), 938.27208943(29) MeV	CODATA 2022
G_N; Planck mass $\sqrt{(\hbar c/G)}$	$6.67430(15) \times 10^{-11}$ m³ kg⁻¹ s⁻²; $1.220890(14) \times 10^{19}$ GeV	CODATA 2022
$\gamma - 1$ (Cassini)	$(2.1 \pm 2.3) \times 10^{-5}$	PDG, tests of gravitational theory, eq. 21.60
$\dot{G}/G$ (lunar laser ranging)	$(-5.0 \pm 9.6) \times 10^{-15}$ per year	same review, eq. 21.56

A few words to read the table. $\alpha(0)$ is the electromagnetic coupling at low energy. M_Z, M_W and M_H are the masses of the Z, W and Higgs particles. m_t is the mass of the top quark, the heaviest. m_e and m_p are the masses of the electron and of the proton. G_N is Newton's gravitational constant. Cassini is a space probe that measured γ with radio signals. Lunar laser ranging measures the distance of the Moon with laser beams. In this way one checks whether G changes in time ($\dot{G}/G$).

2. The direction: the couplings seen from the Planck mass

The hypothesis of this section is standard. The three couplings are carried from M_Z up to the Planck mass. The **renormalisation group** equations of the Standard Model are used. They are the equations that say how a coupling runs when the energy changes. No new particle is assumed below M_P.

The one-loop and two-loop coefficients are built from the particle content. They are checked against the literature

(Machacek and Vaughn). Below 172.56 GeV the top quark is removed with a simplified prescription. In this prescription the left-handed bottom quark counts as half a doublet. It is not a complete electroweak matching. The top Yukawa coupling enters at the threshold. The scheme is MS-bar. Where stated, the SU(5) normalisation is used: $g_1 = \sqrt{(5/3)}\, g'$. The end point is the non-reduced Planck mass ($G = \hbar = c = 1$).

running	$1/\alpha_1$ (SU(5))	$1/\alpha_2$	$1/\alpha_3$	$1/\alpha$_Y (SM)	$R = 1/\alpha_3 \div 1/\alpha_2$
reference: two loops, top threshold	33.101	49.200	52.977	55.168	**1.0768**
one loop	33.334	49.506	52.477	55.557	1.0600
two loops, no top threshold (and y_t(M_Z) = 0.99)	33.043	49.147	52.903	55.072	1.0764
two loops, with the three-loop QCD term	33.101	49.200	52.976	55.168	1.0768
end point: reduced Planck mass	34.159	48.394	51.173	56.932	1.0574
end point: 2 M_P	32.646	49.546	53.753	54.410	1.0849
MSSM above 1 TeV (DR-bar)	17.918	23.784	28.364	29.863	1.1926

The table shows how much R changes with the choices of calculation. The MSSM is the supersymmetric version of the Standard Model. It adds many new particles. DR-bar is another convention, similar to MS-bar.

Cross-check with the literature. Buttazzo, Degrassi, Giardino, Giudice, Sala, Salvio and Strumia (JHEP 12 (2013) 089, eq. 61) made an even more precise calculation. They give the Standard Model couplings at the Planck mass with three-loop running and two-loop thresholds. The couplings are given as functions of M_t, α_s(M_Z) and M_W. We use the same inputs: α_s = 0.1180, M_t = 172.56 GeV, M_W = 80.3692 GeV. Their formulas give $1/\alpha_1 = 33.14$, $1/\alpha_2 = 49.18$ and $1/\alpha_3 = 52.94$. So **R = 1.0766**. Here instead one finds 1.0768 ± 0.0014. The agreement supports the numbers. But it does not show that every omitted term is covered by the estimate below.

Uncertainty. Let us see where the uncertainty on $1/\alpha_3(M_P)$ comes from:

- 0.063 comes from α_s(M_Z);
- 0.001 comes from the three-loop QCD term (it is a truncation estimate: how much the part of the calculation left out weighs);
- 0.001 comes from the top Yukawa coupling (m_t(m_t) ± 1.8 GeV);
- less than 0.001 comes from the threshold and from G_N;
- 0.02 is assigned to the missing three-loop electroweak terms. It is an estimate, not a calculation.

Put together *in quadrature* they give **± 0.066**. In quadrature means: one adds the squares and then takes the square root. It is a working total: errors of the data and estimated truncation terms. It is not a measured standard deviation.

On $1/\alpha_2(M_P)$ the uncertainty is ± 0.008. On R it is **± 0.0014**. The inputs alone, propagated directly on the ratio, would give ± 0.0013. There is a detail. The PDG quotes $1/\hat{\alpha}(M_Z) = 127.930$ together with α_s = 0.1187 ± 0.0017. Here it is used with the world average of α_s, without a joint fit. Finally, the MS-bar or DR-bar convention shifts $1/\alpha_3$ by 0.08. It is a choice of definition, not a measurement error.

3. Where the numbers meet

Sometimes certain numbers of physics seem to meet, that is, to have close values. For each case we ask the two questions. Question 1: do they meet without the hypothesis? Question 2: are there other hypotheses that make them meet?

group	question 1: do they meet without the hypothesis?	question 2: other hypotheses that make them meet
The three couplings at high energy	No. In the Standard Model the pairs cross at 1.1×10^{13}, 2.8×10^{16} and 1.7×10^{14} GeV: three different points.	The MSSM above 1 TeV: the three meet within 0.6 % near 1.3×10^{16} GeV. An exact meeting then requires α_s(M_Z) = 0.1219 (0.1143 for M_SUSY = 10 TeV). The measurement gives 0.1180 ± 0.0009. There are also SO(10) with an intermediate scale and extra dimensions (Dienes, Dudas, Gherghetta): they are not

		computed here.
The band 49–55 of the three couplings at M_P	No. The band is 12 % wide (60 % with the SU(5) normalisation).	Any number in the band will do. There are 736 "simple numbers" between 40 and 60. Of these, 79 lie within 2.3 % of the weak coupling. And 9 lie within twice the working uncertainty of the strong one.
Layers ln(M_P/m) against the couplings at M_P	The electron layers are 51.53. They lie 1.45 below the strong coupling: 22 times its working uncertainty.	Chance. There are 2 layer-coupling pairs within 3 % out of 64. By chance 2.1 are expected.
The proton's gravitational weakness	Yes, by known physics. Given m_p/Λ from QCD, $\alpha_G(p) = (m_p/M_P)^2 = 5.9 \times 10^{-39}$. This number is explained by the running of the strong coupling (Wilczek). The record uses the measured proton mass and does not predict it. A shift of 0.07 in $1/\alpha_3(M_P)$ changes $\alpha_G(p)$ by a factor 0.91, at fixed m_p/Λ.	Not needed.
Large numbers (Dirac, Eddington)	$e^2/(G\, m_e\, m_p) = 2.27 \times 10^{39}$ is arithmetic of α and of the two layers.	Dirac thought $G \propto 1/t$: gravity that decreases with time. It would need $\dot{G}/G$ of the order of 10^{-10} per year. Lunar laser ranging excludes it, by a factor of about 7000. Eddington thought of the integer 137. But $1/\alpha(0) = 137.035999177(21)$ differs from 137 by 0.036. It is about 1.7 million times the measurement uncertainty.

A few explanations. The **layers** are the number ln(M_P/m). They say how many times, in a certain sense, one must "multiply the scale" to go from the mass of a particle to the Planck mass. Λ is the typical energy scale of the strong force. $\alpha_G(p)$ measures how weak gravity is between two protons: 5.9×10^{-39} is a very small number.

The numbers meet only where known physics makes them meet. The historical case is that of the three couplings and supersymmetry. It shows well the logic of the two questions. There are three constants. The meeting costs two parameters: the scale and the common value. One prediction remains: α_s. And the measurement puts this prediction under tension.

4. What R depends on

R is defined with some precise choices. The Standard Model with no new particles below M_P. Two loops. The MS-bar scheme. The non-reduced Planck mass as end point. With other choices R changes:

- at one loop it is 1.0600;
- with the reduced Planck mass it is 1.0574;
- with 2 M_P it is 1.0849;
- with the MSSM it is not defined in the same way (it is 1.1926 at M_P).

The choice of the end point is part of the reading. The model takes the Planck scale as the scale where gravity has one reading per cell. This is the non-reduced Planck mass.

5. The reading proposed by the model

This section uses the language of the MM model. The picture behind it is described in a separate document of the package. Nothing in sections 1–4 depends on this reading.

5.1 Counts

In the model the state of a place is a **register**. A register is a list of T readings divided into K categories. Think of a notebook with K columns, where you mark each event with a cross in its column. The register is updated with Bayes' rule. It is the rule that says how an estimate changes when a new datum arrives. The result is a Dirichlet posterior distribution.

The coupling of a cell is read as the inverse of its number of readings. A new reading arrives. By how much does the state of the register move? The shift is measured with the **Fisher metric**. It is a way of measuring how much two states can be told apart. The expected square of the step is

$$E[\Delta s^2] = (K - 1)/(T + 1)^2,$$

for every composition of the register. It is checked with symbolic calculation for K = 4. T also includes any starting counts (prior pseudo-counts).

There is a limit. The finite Fisher–Rao distance agrees with this formula only when every category has many readings. For a uniform register with T = 4 the exact expected value is 13 % smaller. At T = 1000 the difference is 0.1 %. Near the edge of the simplex many readings in total are not enough. The simplex is the space of all possible compositions. An example: counts (1, 1, 1, 999997), with $T = 10^6$. Here the value is 31 % smaller.

The root-mean-square step per reading therefore goes as $\sqrt{(K - 1)/(T + 1)}$. In words: **at fixed K, T + 1 is inversely proportional to the unit step.** More readings, smaller step. Think of a tub of water. With little water a glass raises the level a lot. With a lot of water it raises it very little.

5.2 Time, space and the two couplings

The reading assigns the weak coupling to the unit of time. It assigns the strong coupling to the unit of space. Now let us hold fixed the ratio between the cell and the unit of time. We express time in the unit of space. This divides the cell by the original ratio between the two units. And, by 5.1, it multiplies T + 1 by the same ratio. The reading identifies the count $1/\alpha$ with T. It is a convention, discussed further below. The data then fix that ratio:

$R = 1.0768 \pm 0.0014$.

What does it mean? At the origin the count in the unit of space is 7.7 % larger than the one in the unit of time. With the step taken as 1/T, which is the convention of this reading, the unit step of space is 7.1 % smaller. An example to understand: it is like measuring the same road with two rulers. If one of the two is a bit shorter, more marks are needed for the same road.

In the model this ratio is an **initial datum**. It is not derived. And it needs no further justification. The identification uses the counts $1/\alpha$. With T + 1 in place of T the ratio of the steps would be 1.0753. The difference is 0.0015, about as large as the uncertainty of R. So the convention is part of the reading.

5.3 The test with gravity

Suppose that R is the original ratio between space and time. Then it must appear wherever a measure of space is compared with a measure of time. Gravity is a good place to check.

In the gravitational sector of the model there are two numbers. The **lapse N** concerns time: it says how slowly a clock runs near a mass. The **conformal factor ψ** concerns space: it says how much rulers stretch. They are tied by $\psi + N\psi = 2$. This rule gives the exterior Schwarzschild metric in isotropic coordinates. It is Einstein's description of the empty space around a spherical mass.

There are three ways of inserting R. They were registered before the calculation.

how R enters	γ	β	against Cassini
R weighs the time part: $\psi + R\,N\psi = 1 + R$	$2R/(1+R) = 1.0370$	$(3R+1)/(2(1+R)) = 1.0185$	1607 × the Cassini error
R weighs the space part: $R\psi + N\psi = 1 + R$	$2/(1+R) = 0.9630$	$(R+3)/(2(1+R)) = 0.9815$	1608 × the Cassini error
R is a ratio of units only	1	1	compatible

The parameters γ and β describe gravity in the Solar System. In Einstein's relativity they are exactly 1. The Cassini probe measured γ with great precision.

Let us also include the working uncertainty of R, with a linear propagation. Then the first two forms lie about 59 times the combined error away. They are two **readings that fall**: R as a physical difference in time, or in space. Of the three forms tested, only the third is compatible with gravity. In this form R is a ratio of units. It is invisible to gravity. It is visible in the couplings, because readings are counted per cell, that is, in the unit.

Pay attention to two things. This does not show that every other physical reading of R is excluded. And the third form is not a new prediction about gravity: it keeps the relation that already gives $\gamma = \beta = 1$.

5.4 What R does not fix

There is a third coupling: the **hypercharge** coupling. It is the part of electromagnetism that the reading calls the **descriptor**. R does not fix it. Eight simple rules were tried. The descriptor read in the unit of time, in that of space, or in both (sum or mean). With the charge unit of the Standard Model or of SU(5). None works. The closest miss is off by 1.3 in $1/\alpha_1$. Hypercharge remains a separate datum.

5.5 What follows

With R and the measured weak coupling, the strong coupling at the Planck mass follows. Then, with the known running, α_s(M_Z) follows.

The gravitational weakness of the proton, 10^{-39}, is linked to this by one route only. It is the QCD relation between the scale Λ and the proton mass, taken from standard physics. The value used in the record is computed from the measured masses m_p and M_P. The record does not predict the proton mass. In the reading, the 10^{-39} is the ratio of the two units at the origin, carried down by the running of the strong coupling.

The number of inputs is the same as in the Standard Model. There is one datum (R) in place of one measured constant (α_s).

6. What is known and what is proposed

The table separates what physics already knows from what this record proposes.

correlation used	in standard physics?
electromagnetism as a mixture of the weak and hypercharge couplings	yes: Glashow 1961, Weinberg 1967, Salam 1968
the strong coupling runs faster than the weak one because of its colours	yes: Gross and Wilczek, Politzer, 1973
the three couplings nearly meet at high energy	yes: Georgi, Quinn, Weinberg, 1974
heavy particles decouple from the running below their mass (in MS-bar this is done with thresholds and matching)	yes: Appelquist and Carazzone, 1975
the weight of a charge is quadratic	yes: Born's rule and quantum electrodynamics
the proton's 10^{-39} comes from the running of the strong coupling	yes: Wilczek, 2001
the electron has three times the charge of the d quark	yes, from anomaly cancellation (Geng and Marshak; Babu and Mohapatra; Minahan, Ramond and Warner, 1989–90)
a ratio between units of space and of time is invisible to gravity	yes: c as a conversion factor
couplings as counts of readings per cell	proposed here
the strong-to-weak ratio at the origin as the ratio between the units of space and of time	proposed here, as far as the author knows
different couplings for the space and time directions	known in another sense: on anisotropic lattices the two couplings belong to the directions of one gauge theory (Karsch 1982; Carena et al. 2022), not to two different forces
the step per reading, inversely proportional to the readings, as the unit of time	information geometry is known (Fisher, Amari); its use as the unit of time is proposed here

A note on the row "c as a conversion factor". The speed of light c converts seconds into metres. This is why changing the ratio between the two units cannot be seen in gravity.

Standard physics gives each correlation its own reason: the symmetry group, the charges, the running. The reading proposes a common interpretation. There is the same original cell. It is counted in the unit of time, in the unit of space or by the descriptor. The reading does not derive the symmetry groups, the list of particles or the charges. These enter through the running. The numbers coincide with those of standard physics. They must coincide, if the two describe the same world.

7. Registered attempts to obtain the strong coupling from the model's counts

Before the calculation, 28 mechanisms were written down and registered. Each was a way of deriving the strong coupling from the model's own counts. They come in four families, generated by fixed rules:

- the entropy A/4 of every surface of the model's minimal saturated place, in nats and in bits (12 mechanisms);
- the exponent between the three dimensions and the two-dimensional mirror of the model (4);
- the model's count 16π, placed at scales near M_P, with the running in between (6);
- 16π plus a small count of the model (6).

The nat and the bit are two units for measuring information, like metres and feet for lengths. Each mechanism was carried to α_s(M_Z) with the reference running.

Result: **0 hits out of 28**. The tolerance was ± 0.0018 on α_s(M_Z). By chance 0.22 hits were expected. That is, there was a probability of 20 % of getting at least one. The closest are 16π + 3 (α_s = 0.1140) and 16π + 2 (0.1292).

For eight candidates ($1/\alpha_3$ below 34) there is no solution for α_s(M_Z) up to 0.9, the largest value tried. For five (A9–A12 and B2, with $1/\alpha_3$ between 85 and 356) no solution was found in the range explored with these equations. The running reaches M_P only for α_s(M_Z) above about 0.0264. And at that edge the top Yukawa coupling at M_P is about 6000. It is far outside the perturbative regime, the one where calculations in small steps work. They are all misses.

The model's count 16π at M_P gives α_s(M_Z) = 0.1765. The proton gravitational coupling that follows is about 40 times larger than the measured value. But it is only indicative. The recipe for Λ is used outside its range ($\Lambda^{(3)}$ above m_c).

Before the registration two expressions had been noticed: 16π + e and 16π at M_P/(4π). They reproduce α_s(M_Z): they give 0.1179 and 0.1163. But they had been found by looking at the target. They are reported as examples of how easy it is to hit a known number a posteriori, that is, after having seen it.

8. What is not claimed

- **No constant is derived.** R is read from the data. Its role as the original ratio between the units is the proposed reading.
- **R depends on the choices of section 4.**
- **In standard physics the couplings have no units** and do not depend on units. The reading assumes that the readings per cell are counted in the unit of the corresponding measure. Let us think about what would happen if the two forces saw different ratios between space and time in their motion. Particles would have different limiting speeds. Coleman and Glashow give two bounds. The first is $1 - c^2 < 10^{-23}$, from cosmic-ray protons, on one side only. The second is $|1 - c^2| < 6 \times 10^{-22}$, from spectroscopy. Here c is the speed of light in units of the maximal speed attainable by matter. For other particles the bounds are much weaker. And none of these bounds concerns R directly. The reading places R in the counting, not in the motion. It is the same conclusion reached with gravity in 5.3.
- **16π is not the count of any coupling at M_P.** 16π is a working value used in the model for the count of the original cell. It lies 1.07 from the weak coupling: 135 times its uncertainty. It lies 2.71 from the strong one: 41 times its working uncertainty.
- **The hypercharge coupling is not fixed by R.**

B. The lepton masses: a balance between the same and the different

Now we move on to the second part: the masses of the particles. Particles come in families of three. For example the electron, the muon and the tau are three *generations* of the same family. They have the same charge but very different masses: the tau weighs more than three thousand times the electron.

The data

The masses of the charged leptons come from CODATA 2022. For the tau: $m_\tau = 1776.93 \pm 0.09$ MeV (PDG).

For the neutrinos we do not know the masses one by one. We know the differences of the squares of the masses, measured with neutrino **oscillations**. A neutrino that travels changes type and then returns to what it was, at regular intervals: this is why one speaks of oscillations. From how much it oscillates these differences are obtained (PDG 2025 lepton summary):

- $\Delta m^2_{21} = (7.50 \pm 0.19) \times 10^{-5}$ eV²;
- $\Delta m^2_{32} = (2.451 \pm 0.026) \times 10^{-3}$ eV² in the normal order, or $(-2.527 \pm 0.034) \times 10^{-3}$ eV² in the inverted order.

Cosmology gives upper bounds on the sum of the neutrino masses, at the 95 % confidence level (PDG 2025, Neutrinos in cosmology, Table 26.2):

- 0.077 eV (Planck + ACT + DESI DR2);
- 0.048 eV (with SPT-3G);
- 0.053 eV (PR4 + ACT + DESI DR1 + Lyman-α).

The electronvolt (eV) is a unit of energy, and therefore of mass. It is very small: the electron has about half a million of them.

The direction (a hypothesis)

In a family of three particles with the same charges there is a **mass term**. It links the left and right components of the three generations. It is written as a 3 × 3 **matrix**: a table of numbers with 3 rows and 3 columns. Its singular values are the masses.

Let A be a Hermitian matrix of amplitudes. Hermitian means that it is symmetric in a precise sense. Its eigenvalues a_i satisfy $a_i^2 = m_i$. That is, each amplitude is a square root of a mass. In the mass basis $A = \mathrm{diag}(a_1, a_2, a_3)$. And an amplitude can have a sign, plus or minus. Careful: this is a construction in the mass basis. It is not a property of every chiral mass matrix.

A splits in one way only into two parts. One part is shared by the three generations: **the same**. The other has zero trace, that is, zero sum on the diagonal: **the different**. One writes $A = (\mathrm{Tr}\, A/3)\, I + A_0$. For every Hermitian A with $\mathrm{Tr}\, A \neq 0$ a formula holds. It uses the Hilbert–Schmidt norm, a way of measuring the *size* of a matrix. The formula was checked with symbolic calculation. It was also checked on 1000 random real symmetric matrices and on 1000 complex Hermitian ones:

$$Q = (m_1 + m_2 + m_3)/(a_1 + a_2 + a_3)^2 = (1/3)\,(1 + |A_0|^2/|A_{id}|^2).$$

In words: Q is the sum of the masses, divided by the square of the sum of their roots. Q is 1/3 plus a piece that measures how much the different weighs compared with the same. If the different weighs as much as the same, Q is exactly 2/3.

Where the numbers meet

family	result
charged leptons	$Q = 0.6666645 \pm 0.0000051$. So the ratio of the different to the same is 0.999997. The exact balance requires $m_\tau = 1776.97$ MeV (measured: 1776.93 ± 0.09).
neutrinos, normal order	one solution, with the negative amplitude on the lightest state: m = 0.00036, 0.00867, 0.05026 eV; sum 0.0593 ± 0.0003 eV
neutrinos, inverted order	one solution, negative amplitude on the lightest state: sum 0.1029 ± 0.0007 eV
quarks (section B.2)	Q = 0.85 (u, c, t), 0.73 (d, s, b) with the PDG masses

For the charged leptons Q is almost exactly 2/3. The same and the different weigh almost the same.

For the neutrinos, with all amplitudes positive there is no solution. The negative sign is not chosen by us. It is the only

case with a solution. And it falls on the lightest state, in both orderings. The oscillations also give a minimum for the sum: 0.0589 eV (normal) and 0.0998 eV (inverted).

Now the bounds from cosmology. The ΛCDM bound of 0.077 eV allows the normal-order solution and excludes the inverted one. But the bounds depend on the cosmological model. The w_0w_aCDM bound in the same table is 0.186 eV and allows both. The two tighter bounds, 0.048 and 0.053 eV, exclude every normal-order spectrum allowed by the oscillations. They exclude even its minimum. So they express a tension between cosmology and neutrino masses as such. They are not a test of this reading.

The two questions:

- **Question 1.** The balance today is a known empirical relation: Koide's relation. It holds with the pole masses to one part in 10^5. With the running masses it holds only to one part in 10^3 (Sumino, JHEP 05 (2009) 075). The running masses are the ones that change with energy, like the couplings. The geometric form of the relation was given by Foot (1994): a 45° angle between the vector of the √m_i and the vector (1, 1, 1). The split into the same and the different is a rewriting of it. It is not a new identity of the masses.
- **Question 2.** Brannen (2006) applied it to neutrinos, with a negative root for the lightest state. He found m = 0.000388, 0.00895, 0.0507 eV, with sum 0.0600 eV. The numbers above are the same relation with the 2025 data (sum 0.0593 eV). They are not identical numbers. Sumino (2009) proposed a family gauge symmetry that explains why the relation holds with pole masses.

The reading proposed by the model

The matrix of a family imports "the same" and "the different" in equal measure: $|A_0| = |A_id|$.

What is imported and cannot be derived? The scale of the family and the orientation of the different part. They count as two of the three masses. What is structural? Their balance. It counts as the third mass.

The reading was stated for families without colour. This restriction was written already knowing that the quarks do not satisfy the relation. For the neutrinos the mechanism that gives them mass is not specified. The calculation uses only the three mass eigenvalues, that is, the three values of the mass.

In the model, mass is the frequency of the internal clock of a particle. This is the pole mass. And the relation holds precisely for pole masses. The registration of this part declares what was known before the calculation.

What is not claimed

No new number. The relation is Koide's. The neutrino prediction is Brannen's, updated. The reading proposes a reason for a known relation. The equal weight of the same and the different remains an interpretive choice. It is not a dynamical explanation. The individual masses are not derived. In the same work there were attempts on the Higgs self-coupling and on the top Yukawa coupling. They were not blind. For this reason they are not reported as results.

B.2 The quarks: the balance in the mirror

Now the quarks. There are two families of quarks. The **up** family: u, c, t. The **down** family: d, s, b. Quarks have one property more than leptons: **colour**. It is not a real colour. It is a name for a kind of charge of the strong force, with three values.

The data

The quark masses come from the PDG 2025 summary table:

- u, d, s in MS-bar at 2 GeV: 2.16 ± 0.07, 4.70 ± 0.07, 93.5 ± 0.8 MeV;
- c and b in MS-bar at their own mass: 1.2730 ± 0.0046 and 4.183 ± 0.007 GeV;
- these five have uncertainties at the 90 % confidence level;
- t from direct measurements: 172.56 ± 0.31 GeV; or in MS-bar: 162.5 +2.1 −1.5 GeV. For this one the PDG does not state the confidence level. Here it is taken as ± 1.8 GeV at one standard deviation.

Quarks are never seen alone. The light quarks have no pole mass. For this reason the balance of a quark family depends on how the masses are defined. So the masses are also carried to one common scale, M_Z. The QCD running of α_s and of the masses is used.

Where the numbers meet

family	Q, PDG masses (t direct)	Q, PDG masses (t MS-bar)	Q, all masses at M_Z
u, c, t	0.8488	0.8449	0.8868
d, s, b	0.7313	0.7313	0.7470

The definition of the masses moves Q by up to 0.042. For the charged leptons the uncertainty on Q is 0.000005. Neither of the two quark families is close to 2/3.

Three rules registered for how colour shifts the balance

Let us write $\rho = |A_0|/|A_{id}|$: the ratio of the different to the same. Then $Q = (1 + \rho^2)/3$. For the leptons $\rho = 1$. Before the calculation three rules were written, starting from the premises of the reading:

1. the different part of a quark family is read in the unit of space, the same part in the unit of time: $\rho = R$ for both families;
2. colour has three directions, and the model's comparison in three dimensions carries the power 3/2: $\rho = R^{(3/2)}$ for both;
3. the weight grows with the readings shared, that is, with the charge in thirds (2 for the up family, 1 for the down family): $\rho = R^2$ and $\rho = R$.

A rule passes if it holds for both families at M_Z. The tolerance is the spread between the definitions plus twice the PDG uncertainties: 0.043 for up, 0.017 for down.

rule	Q predicted, up / down	up family (0.8868)	down family (0.7470)
$\rho = R$	0.7198 / 0.7198	no	no
$\rho = R^{(3/2)}$	0.7495 / 0.7495	no	yes
$\rho = R^2$ (up), R (down)	0.7814 / 0.7198	no	no

None passes. They are three readings that fall. The single hit is what chance allows. With tolerances this wide, let us take a value of Q at random between 0.6 and 0.95. It falls within the tolerance of at least one of the six rules $\rho = R^n$ (n = 1/2, 1, …, 3) in 71 % of cases for up and in 55 % for down. The expected numbers of rules within tolerance are 1.48 and 0.59. The data would require n = 3.43 (up) and 1.46 (down).

Two things were known before and are declared. Before the calculation it had been noticed that the ρ of the up family is close to R^3. With no logic written in advance, that rule was not tested. Then the value $(1 + R^2)/3 \approx 0.720$ had been noticed near the down family, in the registration of Part B. That was before rule (1) was written. So the rules are not independent of the known values.

Two departures from the registration were found afterwards and are declared. First: the registration states two loops for the running of the masses, but the calculation uses three. Second: the registration states two standard deviations of the masses. But the uncertainties of u, d, s, c and b are at the 90 % confidence level. For these masses the tolerance used corresponds to about 3.3 standard deviations. The calculation was redone as registered: two loops; two standard deviations for u, d, s, c, b; the top kept at ± 1.8 GeV. Then Q at M_Z becomes 0.8854 (up) and 0.7463 (down). The tolerances become 0.042 and 0.016. Adding the uncertainty of α_s in the running of the masses they stay 0.042 and 0.016 (0.044 and 0.017 with three loops). **No outcome changes.** The spread between the definitions is a working tolerance, not a statistical interval.

The mirror

Now a different idea: the **mirror**. Take two whole numbers, for example 4 and 6. Their greatest common divisor is 2. Their least common multiple is 12. And $2 \times 12 = 24 = 4 \times 6$. It always holds: greatest common divisor × least common multiple = product. On a logarithmic scale the common multiple is the mirror image of the common divisor. The mirror is at the geometric mean.

A mass read in the unit of the common multiple therefore goes as the inverse mass, 1/m. Two predictions were registered:

- Q(1/m) = 2/3 for both quark families;
- reading a family as m^p, the same exponent p for both families.

family	Q(1/m), PDG masses	Q(1/m), at M_Z	p < 0 with Q(m^p) = 2/3 (PDG / M_Z)	p > 0 (PDG / M_Z)
u, c, t	0.9178	0.9140	−0.514 / −0.512	+0.610 / +0.550
d, s, b	0.6647	0.6674	−1.005 / −0.998	+0.859 / +0.829
e, μ, τ (pole masses)	0.8514	–	−0.651	+1.000

As registered, **neither of the two predictions is satisfied**. The up family does not balance in the mirror. And the two families require different exponents. The down family instead does balance in the mirror:

- let us move d, s and b to the ends of their PDG intervals (90 % confidence level, without correlations): Q(1/m) ranges from 0.6623 to 0.6670, and 2/3 lies inside; this however is not a 90 % interval on Q;
- exactly 2/3 requires m_s/m_d = 20.28; the central values give 19.89; the PDG summary table gives m_s/m_d = 17–22;
- all three masses are needed: without the bottom quark the same expression gives 0.70;
- the exponent required is −1.005 (PDG masses) and −0.998 (at M_Z), compatible with −1, the pure mirror.

There was a third registered check. It weights the three up quarks with 1, R and 2R/(1+R). They are the value of the cell in time, in space and in the part that mirrors itself. It moves Q by less than 0.01, as expected. It does not reach 2/3.

The two questions:

- **Question 1 and question 2.** The mirror balance of the down-type quarks is already in the literature. Rivero (2026) proposes exactly this relation: $(1/m_d + 1/m_s + 1/m_b)/(1/\sqrt{m_d} + 1/\sqrt{m_s} + 1/\sqrt{m_b})^2 = 2/3$. He finds 0.66750 at M_Z (here 0.6674). And he finds exactly 2/3 near 280 TeV, with Standard-Model running. This scale depends on the inputs and on how the running is implemented. The code of this record carries the masses only up to M_Z. The mixed triplets are also known. The triplet (c, b, t) is close to 2/3 (Rodejohann and Zhang 2011; Kartavtsev 2011; Cao 2012, who finds 0.6688). The triplet (s, c, b) is too, with a negative root for s (Rivero 2011). With the PDG masses here they give 0.669 and 0.675. With all masses at M_Z they give 0.720 and 0.697. So they come close to 2/3 with masses defined at different scales. And they move away at the common scale M_Z tried here.

What is proposed and what is not

The picture that comes out is this:

- the charged leptons balance with their direct masses;
- the down-type quarks balance in the mirror;
- the up-type quarks balance in neither of the two ways.

For the down-type quarks the reading proposes a reason: the mirror between the common divisor and the common multiple. The step from whole numbers to the inverse of the real masses is a hypothesis of the reading. In the same way the reading proposes a reason for the leptons. It gives none for the up-type quarks. And it does not say why the leptons are read directly and the down-type quarks in the mirror. No rule linking leptons and quarks was found. The relation of the down-type quarks is not new.

How the work was done

The author. The author is an independent researcher, not a physicist. The calculations and the text were produced with AI assistants. Every computed number comes from the code of this record.

The registrations. The registrations were written before the calculations they concern. The folder `records/` holds the five registrations written before the calculations of sections 5.3, 7, B and B.2. They are in Italian, with their SHA-256

fingerprints and the local times (`SHA256SUMS_REGISTRAZIONE.txt`, `SHA256SUMS_REGISTRAZIONE_MASSE.txt`, `SHA256SUMS_REGISTRAZIONE_QUARK.txt`). A fingerprint is a code computed from a file: if the file changes even by a comma, the fingerprint changes. The times are recorded by the author's machine. They are not an external timestamp. The file `records/README.md` summarises them in English. In particular:

- the three ways of inserting R into gravity (section 5.3) are in `records/REGISTRAZIONE_R_ALTROVE.md`;
- the 28 mechanisms of section 7 are in `records/REGISTRAZIONE_FORTE_MM.md`, SHA-256 37c3ac7b…, time recorded locally 27 Sep 2026 11:40:19;
- Part B is in `records/RIFINITURA_E_REGISTRAZIONE.md`, SHA-256 ec65f7c6…, local time 27 Sep 2026 14:08:45;
- the three rules on the quarks are in `records/REGISTRAZIONE_QUARK.md`, the mirror in `records/REGISTRAZIONE_QUARK_SPECCHIO.md`.

The sources and the results. The sources of the data and their fingerprints are in `sources/SOURCES.md`. The full data table is `results/DATA_TABLE.md`. The picture of the model used in section 5 is described in `WHERE_THIS_COMES_FROM.md`. The result files cited in the original text are: `results/uncertainty.log` (uncertainties, section 2), `code/literature_check.py` (check against Buttazzo and colleagues), `results/meetings.log` (meetings between numbers, section 3), `results/reading.log` (step per reading, section 5.1), `results/data.log` (gravitational weakness of the proton, section 5.5), `results/strong_from_counts.log` (the 28 mechanisms, section 7), `results/masses.log` (the formula for Q, Part B), `code/quarks.py` and `results/quarks.log` (quarks, section B.2; the departures from the registration are in part 7 of the log).

How to redo the calculations. You need Python 3.10 or later, with numpy, scipy and sympy. From the folder `code/` give the command:

python run_all.py

The script runs, in this order, `data.py`, `running.py`, `literature_check.py`, `uncertainty.py`, `meetings.py`, `strong_from_counts.py`, `reading.py`, `masses.py` and `quarks.py`. It writes their logs and JSON files in `results/`. It lists the SHA-256 fingerprints of code and results in `results/SHA256SUMS_run.txt`. It takes a few minutes on a four-core desktop computer.

This version. This student version was written with an AI assistant (Claude) from the archived text in English. First a complete Italian translation was made, then the rewriting for a school student. The numbers were checked with a program against the archived text. This English text is a translation of the Italian student version, which the author read and approved.

Glossary

- **Balance between the same and the different.** The reading of Part B: the part that the three generations share and the part that distinguishes them weigh the same. It is Koide's relation.
- **Colour.** A charge of the quarks for the strong force, with three values. It is not a real colour.
- **Coupling (α).** The number that says how intense a force is. Its inverse, 1/α, is large when the force is weak.
- **Descriptor.** In the model, the part of the description that makes an interaction electromagnetic. Here it corresponds to the hypercharge coupling.
- **Layers.** ln(M_P/m): the number of e-folds of scale between the Planck mass and the mass of a particle.
- **Leptons.** A group of particles: electron, muon, tau and the three neutrinos. The three with the same charge form a family.
- **Loops.** The degrees of precision of the calculation of the running. Two loops are more precise than one.
- **Minimal saturated place.** In the model, the smallest place whose count is saturated. Its entropy A/4 is 16π in the model's units.
- **Mirror.** The idea that the common multiple is the mirror image of the common divisor. It leads to reading the inverse masses, 1/m.
- **MS-bar, DR-bar.** Conventions for defining quantities in calculations, like always measuring height without shoes.
- **Planck mass (M_P).** An enormous energy, where gravity becomes as strong as the other forces.
- **Pole mass.** The mass of a free particle. The light quarks do not have one, because they are never seen alone.
- **R.** The ratio between the inverse of the strong coupling and that of the weak one at the Planck mass: 1.0768 ± 0.0014. In the reading it is the original ratio between the unit of space and the unit of time.
- **Reading, readings per cell.** In the model, the elementary event by which a place registers something. The number of readings in the elementary cell is read as the inverse of a coupling.
- **Register.** In the model, a list of readings divided into categories, like a notebook with columns.

- **Registration.** A prediction written and fixed with a fingerprint before the calculation, so that it cannot be changed afterwards.
- **Running of the couplings.** The fact that a coupling changes with the energy at which it is observed.
- **Standard deviation.** The typical measure of how much a result can vary because of random errors.
- **Standard Model.** The theory of the particles and of the three forces between them: strong, weak and electromagnetic.
- **Unit of time, unit of space.** The elementary step of the cell, measured as time or as space.
- **Up and down quarks.** The two families of quarks: u, c, t and d, s, b.

References

- S. Navas et al. (Particle Data Group), Phys. Rev. D 110, 030001 (2024) and 2025 update.
- P. J. Mohr, D. B. Newell, B. N. Taylor, E. Tiesinga, CODATA 2022, Rev. Mod. Phys. 97, 025002 (2025).
- M. E. Machacek, M. T. Vaughn, Nucl. Phys. B 222, 83 (1983); B 236, 221 (1984); B 249, 70 (1985).
- S. P. Martin, A supersymmetry primer, hep-ph/9709356.
- D. J. Gross, F. Wilczek, Phys. Rev. Lett. 30, 1343 (1973); H. D. Politzer, Phys. Rev. Lett. 30, 1346 (1973).
- H. Georgi, H. R. Quinn, S. Weinberg, Phys. Rev. Lett. 33, 451 (1974).
- T. Appelquist, J. Carazzone, Phys. Rev. D 11, 2856 (1975).
- F. Wilczek, Scaling Mount Planck I: a view from the bottom, Physics Today 54 (6), 12 (2001).
- D. Buttazzo, G. Degrassi, P. P. Giardino, G. F. Giudice, F. Sala, A. Salvio, A. Strumia, Investigating the near-criticality of the Higgs boson, JHEP 12 (2013) 089, arXiv:1307.3536.
- C. Q. Geng, R. E. Marshak, Phys. Rev. D 39, 693 (1989); K. S. Babu, R. N. Mohapatra, Phys. Rev. Lett. 63, 938 (1989); J. A. Minahan, P. Ramond, R. C. Warner, Phys. Rev. D 41, 715 (1990).
- K. R. Dienes, E. Dudas, T. Gherghetta, Phys. Lett. B 436, 55 (1998).
- Y. Koide, Lett. Nuovo Cim. 34, 201 (1982); Phys. Lett. B 120, 161 (1983).
- C. Brannen, Koide's mass formula for neutrinos, APS Northwest Section Meeting (2006); the values quoted are from his talk «Koide formula for neutrino masses», Joint Meeting of Pacific Region Particle Physics Communities, Waikiki, 1 November 2006.
- R. Foot, A note on Koide's lepton mass relation, hep-ph/9402242 (1994).
- Y. Sumino, Family gauge symmetry as an origin of Koide's mass formula and charged lepton spectrum, JHEP 05 (2009) 075, arXiv:0812.2103.
- W. Rodejohann, H. Zhang, Extended empirical fermion mass relation, Phys. Lett. B 698, 152 (2011), arXiv:1101.5525.
- A. Kartavtsev, A remark on the Koide relation for quarks, arXiv:1111.0480 (2011).
- A. Rivero, A new Koide tuple: strange-charm-bottom, arXiv:1111.7232 (2011).
- F. G. Cao, Neutrino masses from lepton and quark mass relations and neutrino oscillations, arXiv:1205.4068 (2012).
- A. Rivero, New sum rules of the Koide type, arXiv:2606.10060 (2026).
- S. Coleman, S. L. Glashow, High-energy tests of Lorentz invariance, Phys. Rev. D 59, 116008 (1999), arXiv:hep-ph/9812418.
- F. Karsch, Nucl. Phys. B 205, 285 (1982); M. Carena, E. J. Gustafson, H. Lamm, Y.-Y. Li, W. Liu, Gauge theory couplings on anisotropic lattices, Phys. Rev. D 106, 114504 (2022).
- B. Bertotti, L. Iess, P. Tortora, Nature 425, 374 (2003).
- P. A. M. Dirac, Nature 139, 323 (1937).
- S. Amari, Information Geometry and Its Applications, Springer (2016).
- T. Jacobson, Phys. Rev. Lett. 75, 1260 (1995); E. Verlinde, SciPost Phys. 2, 016 (2017): two other approaches in which gravity is read from information.
- R. Bousso, J. Polchinski, JHEP 06 (2000) 006; L. Susskind, hep-th/0302219: the view in which constants are environmental and not derivable.